

Outline

Part 1: Basics

- Denoising diffusion probabilistic models (DDPMs)
- Conditional diffusion models
- Large-scale models: DALL-E 2, Stable Diffusion, Imagen

Part 2: Recent Advances

- Denoising diffusion implicit models (DDIMs)

Denoising Diffusion Implicit Models (DDIMs)

- DDIM **roughly sketches** the final sample, then **refine** it with the reverse process
 - **Key idea:**
 - Given $\mathbf{x}_t$, generate the **rough sketch** $\mathbf{x}_0$ and **refine** $p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)^1$
 - Unlike original diffusion model, it is not a Markovian structure

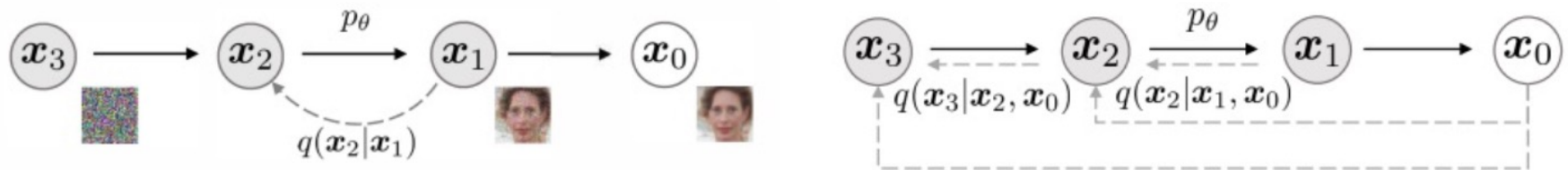


Figure 1: Graphical models for diffusion (left) and non-Markovian (right) inference models.

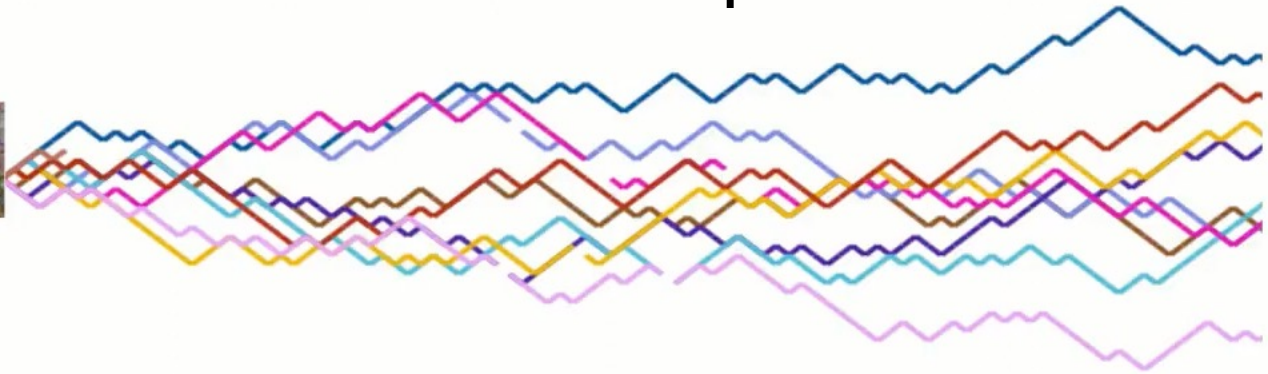
DDIM follows a deterministic process

DDPM follows stochastic process

DDPM



x_T



DDIM follows deterministic process

DDIM



x_T



The same original noise x_T to same image x_0

[Tidiane Ndir, presentation on Faster diffusion, Uni of Freiburg](#)

DDIM reduces the sampling steps significantly

- Creates the outline of the sample after only 10 steps (DDPM needs hundreds)

